

ISEP — Digital Engineering School

Compensation project 2025/2026 · 5 ECTS · due 31 August 2026

Gym Buddies

An individual web application that connects athletes so they can train together, motivate each other, and find a gym buddy.

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This report **summarizes** the wiki in gym-buddy-documentation (commit **2fcabfa**). It is not a second specification. Functional and technical rules live in that repository.

1. Introduction and academic framing

The assignment covers Software Engineering, Web Technologies, and Algorithms and Advanced Programming. Work is individual. Defense is 20 minutes plus 30 minutes of questions. The instructor account `maurras.togbe@isep.fr` must be a collaborator on every private repository.

There was no instructor cadrage meeting (instructor on holiday, recorded 2026-08-19). The brief does not require wiki pages to stay Draft until that meeting. Implementation followed wiki, then the OpenAPI tag, then generate, then implement.

Repositories

Repository	Role	URL
gym-buddy-documentation	Wiki, tickets, Gym Buddy Project	github.com/.../gym-buddy-documentation
gym-buddy-openapi	OpenAPI 3.1 \$ref tree (HTTP source of truth)	github.com/.../gym-buddy-openapi
gym-buddy-service	Java 25 LTS / Spring Boot API	github.com/.../gym-buddy-service
gym-buddy-ui	Angular 22 member app + /admin	github.com/.../gym-buddy-ui

2. Problem and users

Athletes who want a training partner have no product that combines a friends graph, public/private profiles, capacity-limited sessions, and explainable “why this person” suggestions. Gym Buddies is that product for members; staff get a separate back-office bundle. Native mobile, payments, and wearables are out of scope.

3. Functional overview

Area	FS prefix	On develop
Accounts / JWT	FS-ACCT	Register, login, refresh, logout, password, close
Profiles	FS-PROF	Public / private, stub for strangers
Friends	FS-FRND	Request, accept, block
Feed / posts / comments	FS-FEED / POST / CMT	Friends feed, likes, depth 4
Events	FS-EVT	Instant / WEEKLY, apply, transactional capacity
Search / suggestions	FS-SRCH / SUGG	People+events; FoF generate-and-score
Messaging / media	FS-MSG / MED	Text, image, audio; MinIO signed URLs
Admin / fixtures	FS-ADM	Roles, hide, audit; Datafaker seed 20260813

Highlights for the defense: the friends news feed (FS-FEED), friends-only events with a last-seat CONFLICT (FS-EVT-07), and suggestion cards that print a primary reason (FS-SUGG-03).

4. Architecture and data model

Gym Buddies is a **modular monolith** behind one HTTP API and a WebSocket gateway. Two Angular clients (member and /admin) share that API. Bounded contexts (auth, social, events, search, chat, media, admin) live in one deployable. Microservices were rejected: operational cost dwarfs the academic benefit for one student.

Runtime: PostgreSQL 18 (system of record), Redis (refresh denylist, suggestion cache), MinIO (images and audio — never an API /uploads directory). The UI is static on GitHub Pages; the API is a Docker image on an OVH VPS behind Caddy, bound to 127.0.0.1:8080.

Core entities: User, Profile, Friendship, Post, Comment, Event (plus occurrences and applications), Media, Conversation/Message, AuditEvent. Identifiers are UUIDs. Timestamps are UTC. Soft-delete keeps nested threads and moderation consistent. The wiki ER diagram is the authoritative picture (20-Architecture/06-Data-model.md).

5. Algorithms

5.1 Friend suggestions

Two-stage generate-and-score. Candidates (at most 200): friends of friends, same city and sport, co-participants in the last 90 days; minus self, friends, pending, blocked, dismissed, locked. Score $S(u,v) = 0.35 m + 0.25 J + 0.15 G + 0.15 T + 0.10 E$, with m = Adamic-Adar on mutual friends, J = Jaccard of sports, G = geo ($D = 25$ km), T = window overlap, E = experience closeness. The card primary reason is argmax of weight times feature. FoF is $O(d^2)$ per user; nightly all-users is fine at 3 000 users. Collaborative filtering was rejected: no implicit-feedback volume, and it cannot explain why.

5.2 Filtered search

People and events are two indexes with cursor pagination. Filters include sports, city, radius, remaining capacity, and friend-state. Private strangers never appear. Default sort is relevance then recency. PostgreSQL is enough at this scale; Elasticsearch was rejected for MVP.

5.3 User matching

Weekly opt-in greedy assignment with a unique pair, no block edges, and a draft instant event of capacity 1 (visibility friends). Greedy is a 1/2-approximation; the wiki records that we do not show an empirical gap versus exact. Event accept order (FS-EVT-13) reuses the same matching score as a suggested queue, not as a capacity override.

6. Security

Access JWT is HS256 in JSON. Refresh is a cookie (HttpOnly, Secure, SameSite=Lax, path /api/v1/auth) rotated on use and denylisted in Redis on logout. Passwords are Argon2id, never logged. Missing ACL returns NOT_FOUND (no existence leak) unless a spec names FORBIDDEN. Every file download is a 60-second signed GET minted only after canRead. Member calls to /admin/* return NOT_FOUND. Staff JavaScript is a separate Angular bundle.

7. Implementation notes

OpenAPI 3.1 in gym-buddy-openapi is the HTTP source of truth. The service generates Java models and API interfaces at build (controllers implement those interfaces; generated sources are not committed). The UI generates a TypeScript client with orval from node_modules/gym-buddy-openapi/openapi/openapi.yaml. Spring springdoc /v3/api-docs is never the published contract.

Gitflow: feature branches from develop, PRs to develop, Conventional Commits with the documentation ticket in the scope (feat(#59): ...) and Refs: .../gym-buddy-documentation#N. GitHub Actions: format, tests, HTTP smoke on every PR. Release squash-merges develop onto main, tags vX.Y.Z, and deploys. Application versions stay 0.1.x; 1.0.0 is the academic ship and is never chosen automatically. Live Pages is still a 0.1.1 bundle; the full product is on develop and needs a Release to appear on github.io.

8. Tests and fixtures

Backend unit tests name FS IDs (fsEvt07_concurrentLastSeatAcceptsExactlyOne, fsProf04_strangerOnPrivateProfileSeesStub, ...). Integration tests use Testcontainers. CI smoke hits GET /api/v1/healthz. Fixtures: Datafaker, seed 20260813, default 3 000 users / 12 000 accepted friendships / 15 000 posts. CI uses tens of rows (FixtureMagnitude.tiny()). Spring profile prod cannot reset. About ten stock MinIO objects are reused so the 5 000 media metadata rows do not fill the disk.

9. Critical analysis

Strengths: a demoable goal; a modular monolith that still shows Software Engineering structure; two distinct algorithms (explainable scoring vs constrained assignment); object storage that answers the brief's disk and ACL items; spec IDs that tests cite.

After implementation, the main gaps are honest: suggestion matching is greedy, not exact; DMs are not E2E encrypted (staff can read plaintext); search on messy city strings has no geocoder; GitHub Pages cannot host Java, and UFW plus SameSite=Lax means login-from-Pages is not claimed; live Pages is an older 0.1.1 tag. With two weeks less I would drop weekly matching, audio DMs, and recurrence beyond WEEKLY+UNTIL.

10. Conclusion

Software Engineering is evidenced by UML, Flyway, Gitflow, tickets on one board, and reviews against FS IDs. Web Technologies is evidenced by a versioned OpenAPI contract, JWT, WebSocket, Angular, and signed media. Algorithms is evidenced by FoF generate-and-score, filtered search, and weekly matching — implemented in Java, unit-tested, without a hidden SaaS.

11. Appendix

Screenshot checklist (03-Screenshots.md)

Shots 1–16 must be captured from the **live Angular UI** after a Release of current develop (or local compose + fixtures). They are **not** in this PDF: the ticket forbids pasting wiki mockup JPGs as if they were product screens, and github.io still serves Pages v0.1.1 (auth-era). Operator: sign in as demo.alex / demo.blake, store files under 99-Academic-deliverables/screenshots/, one-sentence caption each.

#	Shot	FS
1	Register / login	FS-ACCT
2–3	Public profile; private stub	FS-PROF
4	Friend request	FS-FRND
5–7	Feed, likes, nested comments	FS-FEED / POST / CMT
8–10	Event create, accept, recurrence	FS-EVT
11–12	Search; suggestion why	FS-SRCH / SUGG
13–14	Chat; denied media	FS-MSG / MED
15–16	Admin role + hidden post	FS-ADM

UML (use case, activity, sequence, class) is in 60-UML-diagrams. Selected HTTP surface is 40-Technical-specifications/09-Target-HTTP-surface.md. No cadrage minutes were invented.

Formalities: email maurras.togbe@isep.fr by 31 August 2026. Working language of the wiki is English; translate if the instructor requires French. Cite wiki commit 2fcabfa.